CMPS 2200 Recitation 07

Answers

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Place all written answers from `recitation-07.md` here for easier grading.

- **2) (2 pts)** Write a recurrence for the **work** of this algorithm and solve it. Assume the input is n to compute F_n.

  - $W(n) = W(n-1) + W(n-2) + O(1)$

  Costs at level 0: 1

  Costs at level 1: 2

  Costs at level 2: 4

  Cost at level i: 2^i

  The maximum height of the tree is n

  So the work is: $W(n) \leq \sum_{i=0}^{n} 2^i \in O(2^n)$

- **3) (2 pts)** Write a recurrence for the **span** of this algorithm and solve it.

  Assuming the two recursive calls run in parallel, only the longer branch contributes to the span:

  $$S(n) = \max(S(n-1), S(n-2)) + O(1)$$

  Since the $n-1$ branch is longer, this simplifies to

  $$S(n) = S(n-1) + O(1).$$

  The critical path has height n, so

  $$S(n) = \sum_{i=1}^{n} O(1) = O(n).$$

  Therefore, the span is $\Theta(n)$.

- **4) (1 pt)** Inspecting the `counts` list, what interesting pattern emerges?

  Most of the recursive calls is on the first two fibonacci number.

- **6) (2 pts)** When computing F_n, what is the maximum number of times that `fib_top_down(i)` will be called for any value i? Based on this, what is the **work** and **span** of this algorithm?

- **8) (2 pts)** When computing F_n, what is the maximum number of times that F_i will be read from the list for any value i? Based on this, what is the **work** and **span** of `fib_bottom_up`?